Interaction of Surface Type, Temperature, and Week of Season on Concussion Risk in the National Football League

A Bayesian Analysis

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Background: Artificial turf fields and environmental conditions may influence sports concussion risk, but existing research is limited by uncontrolled confounding factors, limited sample size, and the assumption that risk factors are independent of one another. The purpose of this study was to examine how playing surface, time of season, and game temperature relate to diagnosed concussion risk in the National Football League (NFL).

Methods: This retrospective cohort study examined data from the 2012 to the 2019 NFL regular season. We fit Bayesian negative binomial regression models to relate how playing surface, game temperature, and week of the season independently related to diagnosed concussion risk and any interactions among these factors.

Results: We identified 1096 diagnosed concussions in 1830 games. There was a >99% probability that concussion risk was reduced on grass surface (median incidence rate ratio [IRR] = 0.78 [95% credible interval: 0.68, 0.89]), >99% probability that concussion risk was lower at higher temperatures (IRR = 0.85 [0.76,0.95] for each 7.9 °C), and >91% probability that concussion risk increased with each week of the season (IRR = 1.02 [1.00,1.04]). There was an >84% probability for a surface × temperature interaction (IRR = 1.01 [0.96, 1.28]) and >75% probability for a surface × week interaction (IRR = 1.02 [0.99, 1.05]).

Conclusions: Diagnosed concussion risk is increased on artificial turf compared with natural grass, and this is exacerbated in cold weather and, independently, later in the season. The complex interplay between these factors necessitates accounting for multiple factors and their interactions when investigating sports injury risk factors and devising mitigation methods.

Keywords : Athletes; Bayes theorem; Brain concussion; Football; Risk factors; Sports injuries; Temperature

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Concerns regarding neurodegenerative diseases have prompted sports organizations to consider implementing policies and interventions aimed at reducing sport-related concussions.[1] Identifying modifiable risk factors for concussions and repetitive head impacts in American football is particularly important[2] due to the potential risk of chronic traumatic encephalopathy.[3,4]

There has been considerable interest in examining the role of playing surfaces on brain injuries, and artificial turf fields are commonly marketed with claims of reducing the risk of concussions.[5–8] Conversely, the National Football League's (NFL) Players Association (NFLPA) claims that artificial turf is detrimental to player safety and long-term health,[9] but there is currently limited evidence to support this claim specifically with regard to brain injuries. An analysis of the 2012–2013 seasons of NFL data found no association between playing surface and concussion risk.[2] A 2019 systematic review and meta-analysis concluded that concussion risk was slightly decreased when contact sports (soccer, American football, and rugby union) were played on an artificial surface compared with natural grass.[10] The two included studies for American football found no differences in concussion risk between one specific brand of artificial turf product and natural grass over 3 years of college football,[11] and a greater risk of concussion on natural grass over five seasons of high school football.[12]

There are also within-season and temperature trends in injury rates in football and similar sports. Examination of 4 years of NFL data revealed a substantial increase in concussion

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The source code and full dataset used for this study are attached as Supplemental Digital Content (SDC).

SDC Supplemental digital content is available through direct URL citations in the HTML and PDF versions of this article (www.epidem.com).

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risk during the second half of the season, compared with the first half,[13] and for games played in colder weather.[14] Due to the timing of the season, which begins in late summer and ends in early January, and the primarily outdoor nature of the sport, these two factors are substantially correlated and should be considered together using methods to disentangle their associations. Surface may also interact with these variables. While biologic factors, such as a reduced concussion threshold (i.e., the magnitude of injurious stimulus needed to produce concussion symptoms, which may be dependent on dynamic neuroanatomic and physiological factors and thus vary between and within athletes during a season),[15–18] could explain within-season changes in injury risk, it is also possible that within-season changes in the material properties of the playing surface (e.g., friction and shock absorption) could influence concussion risk.[19] These may be due to wear or damage to the field[20–22] or temperature-dependent effects,[23,24] both of which may differ between playing surfaces. It is also possible that these apparent risk factors may not be purely physical or biological but may also reflect bias in suspecting, reporting, or diagnosing concussions.

Playing surface, time of season, and game temperature all potentially influence concussion risk in football, but these factors are interrelated, and existing analyses do not disentangle their potential confounding effects upon one another. Additionally, many analyses of these factors have divided temperature and week of season into categories,[2,13,14] and it is possible that arbitrary categorization could lead to spurious associations for concussion risk.[25,26] Thus, the objective of this study was to holistically examine how playing surface, game temperature, and week of season influenced diagnosed concussion risk in NFL games.

METHODS

Study Design and Data Collection

This research was reviewed by the High Point University Institutional Review Board and was approved under an exempt review (protocol #201607-500).

We performed a retrospective cohort analysis of eight NFL seasons (2012–13 through 2019–20) of diagnosed concussion data from the Public Broadcasting Service Frontline Concussion Watch[27] and Football Outsiders[28] injury databases. The Football Outsiders database includes data from all weekly NFL injury reports during this time period, including concussion and nonconcussion injuries from weeks 1 to 16 of the regular season. To strengthen completeness, we combined this concussion dataset with data from the Frontline database, which provided an independently collected list of concussions incurred by an NFL player from 2012 to 2015. We excluded previously identified erroneous data points within the Frontline data set.[25] We analyzed injury report data from the first 16 weeks of the season (15 games per team), since injury report data from week 17 are not available for teams that did not qualify for the playoffs. This is consistent with previous studies.[25] We assumed that all concussions occurred in a game setting unless (1) previous data from the Frontline database[25,27] specified it occurred during a practice (in which case, it was excluded from analysis) or (2) if a concussion was listed for a player in the absence of him taking any snaps in games that week. Multiple analyses suggest that approximately 95% of diagnosed concussions in the NFL occur during games.[25,29] However, this differs from youth, high school, and college football, where concussions are more likely during practices,[30,31] which is most likely due to the less frequent and more highly regulated nature of contact practices in the NFL during our study's timeframe. We also assume that all of these concussions were diagnosed by appropriately trained medical personnel and therefore consider all of the injuries in this analysis to be "diagnosed concussions" (heretofore, referred to as or used interchangeably with "concussions" unless otherwise specified).

Stadium and Playing Surface

The playing surface was categorized into natural grass, artificial turf, and hybrid (a surface that includes an artificial turf foundation, combined with natural grass, potentially resulting in a unique profile of material properties). Preliminary analysis revealed hybrid surfaces may be associated with a distinct difference in concussion risk compared with natural grass and artificial turf (eAppendix: eTable 1; http://links.lww.com/EDE/C63 and eFigure 1; http://links.lww.com/EDE/C63), but represented <5% of total games (Table 1). Thus, we excluded games played on hybrid surfaces (n = 82) from analysis.

We accounted for aberrations from a team's normal home stadium in the dataset, including games played in international locations and games that were moved to a different stadium due to extenuating circumstances. Likewise, we accounted for within-season and between-season changes in playing surface type within a stadium (e.g., the Houston Texans played one game on natural grass before switching to artificial turf for the remainder of the 2015 season).

TABLE 1. Summary of NFL games included in the primary model. Note, games that were played in stadia with a retractable roof closed, but doors and windows open (n = 8) are excluded from the table

	Indoor Games (Concussions)	Outdoor Games (Concussions)	Combined Games (Concussions)
Artificial turf	359 (218)	441 (334)	800 (552)
Natural grass	63 (25)	967 (519)	1030 (544)
Hybrid surface	0 (N/A)	82 (52)	82 (52)
Combined (including hybrid)	422 (243)	1490 (905)	1912 (1148)
Combined (excluding hybrid)	422 (243)	1408 (853)	1830 (1096)

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Game Temperature

The Pro-Football-Reference database[32] provides official weather details for most games (n = 1557 games). For outdoor games where any of this data was missing, we retrieved temperature from official NFL game reports and annual team media guides (n = 247). If the temperature was not provided in these sources (n = 108 games), data were retrieved from WeatherUnderground[33] from the zip code of the stadium at the nearest time point before kickoff (generally within 1 hour). The official NFL summary occasionally provided the temperature for indoor games (generally 20–22 °C), and when this was not provided, it was assumed to be 20 °C.

To determine the game temperature in stadia with a retractable roof, we examined the respective team's media guide to determine if the game was played under indoor conditions (i.e., roof closed, doors/windows closed) or outdoor conditions (i.e., roof open). If this information was not available in the media guide, we used Wikipedia game summaries, the official NFL summary, news records, and game videos to determine whether the roof was open or closed. We excluded games in which the doors or windows were open but the roof was closed (n = 8), from analyses, because the game temperature was not certain.

DATA ANALYSIS

Primary Model

Data analysis was performed in R Studio v1.4.1106.[34] The data set and source code are available for download http://links.lww.com/EDE/C61; http://links.lww.com/EDE/C62, respectively. Preliminary analysis of the complete dataset revealed the number of concussions within a game is well-represented by a negative binomial distribution (eFigure 2; http://links.lww.com/EDE/C63). Thus, negative binomial regression models were developed to determine the relationship between risk factors and the number of concussions within a game. The number of concussions in a game served as the dependent variable. Week of season (henceforth referred to as "week"), playing surface (artificial turf vs. natural grass), and game temperature served as independent predictor variables.

As noted previously, surface wear occurs over the course of a season, and this may differ between surface types. The material properties of the playing surface also seem to be temperature-dependent, and this may also vary by surface type. Thus, the two-way interactions of surface × week and surface × temperature were included in the model. We did not have any a priori justification to include the two-way interaction of week × temperature or the three-way interaction between all independent predictors in the model. However, we did fit these models in a supplementary analysis (eAppendix: eTable 2A,B; http://links.lww.com/EDE/C63) and found similar results.

The model intercept represents the log of the average number of concussions in a game played on artificial turf in the middle of the season (between weeks 8 and 9) and at an average temperature (17.1 °C). Previous research suggests that, over a 12-year span, the average number of concussions in an NFL was about 0.40.[35] We therefore specified the prior distribution on the model intercept to be normal with a mean of −0.92 (ln 0.4 = −0.92) and with a standard deviation of 0.3. This choice represents a strong prior belief that, in average conditions, there is often 0 or 1 concussions per game.

Previous research using Poisson regression models did not find any relationship between the time of season (four categories of 4 weeks) or playing surface (natural grass vs. artificial turf) across two NFL seasons.[2] Thus, the priors for the model parameters corresponding to "week" and "playing surface" were set to 0 with a normal distribution with a standard deviation of 2.5. For simplicity, we specified identical normal priors for the remaining model parameters. Although these choices are somewhat informative, prior predictive checks[36] revealed that the priors on the parameters induced a fairly noninformative prior on the average number of concussions per game. To wit, the induced prior on the average number of concussions per game placed 50% probability on the range (0.64, 4.39), 80% probability on the range (0.36, 20.23), 90% probability on the range (0.27, 69.14), and 95% probability on the range (0.21, 260.90). Importantly, the induced prior does not concentrate on extremely small values near zero (which would suggest that there are essentially no concussions) or on implausibly large values (which would suggest that there are at least tens or hundreds of concussions per game). Instead, the induced prior covers a broad range, indicating that the prior regularization on the model parameters is not so strong as to bias our model parameters towards one set of conclusions.

We use the posterior mean as the point estimate of each model parameter and report the 95% central credible interval as a measure of uncertainty about these parameters. Since the posterior distribution is analytically intractable, we rely on Markov Chain Monte Carlo (MCMC) simulation to approximate these quantities. Our simulation was performed using the rstanarm package. We specifically ran eight chains for 15,000 iterations each and discarded the first 5,000 iterations as a warmup and credible intervals for all model parameters.

We exponentiated each sample of the model parameters to compute the incidence rate ratio (IRR) of all of the predictive variables in our model. We performed several checks on the reasonableness of our fitted model (see eAppendix, http://links.lww.com/EDE/C63).[36,37]

The median value and 95% central credible intervals were computed for the IRR of each parameter.[38] We additionally computed the posterior probability that the IRR exceeded 1.0 (which would suggest a harmful effect), or was below 1.0 (which would suggest a beneficial effect).

Sensitivity Analyses

To ensure the correlation between week and temperature did not cause collinearity problems for our primary model, we used separate models with novel control methods for each variable. We then compared the coefficients from these models to those of the primary model.

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To determine if the week of season was truly an independent predictor of concussion risk, we ran a model on only indoor games to control for temperature without including it in the model. This allowed us to control for temperature (i.e., it would be similar for all games), without including it as a model parameter, and determine if the week was still an independent predictor of concussion. We ran a similar model using both indoor and outdoor games played within a narrow temperature range (17.8–25.6 °C).

To determine if temperature was an independent predictor, we ran a model on only games during weeks 13–16 without a term for the week in the model. We would not expect a major change in concussion risk from a week alone during this short 4-week span, but games during this time have a wide range of temperatures. Thus, this late-season model would allow us to determine if temperature was still a predictor of concussion within this short time span.

RESULTS

Games Included

There were 1920 games played during the first 16 weeks of the 2012–2019 seasons (Table 1). Of these, a total of 1830 games met the inclusion criteria described in the methods and were included in the primary model. The median temperature was similar between surfaces (~20 °C for both), but temperature distribution varied by playing surface and week of season. Games played on natural grass (n = 1030) followed a left-skewed distribution, whereas games played on artificial turf (n = 800) had a spike at 20 °C (n = 359 indoor games at 20 °C, with n = 441 outdoor games with median 13.9 °C) (eFigure 3A,B; http://links.lww.com/EDE/C63). Median temperature decreased over the course of the season (eFigure 3C; http://links.lww.com/EDE/C63.

Model Results

Model Diagnostics and Fit

Our diagnostics did not reveal problems with convergence or mixing or the simulated Markov chains: all of the R-hat statistics were close to 1.0 (consistent with convergence to the stationary distribution) and visual inspection of the traceplots did not reveal issues with mixing. Additional checks revealed that there was no single game exerting undue influence on our posterior distribution. Detailed discussions of our model diagnostics are available in the eAppendix.

Model Parameters

A summary of model parameters is presented in Table 2, and probability density plots for the posterior distribution with IRR credible intervals for each parameter are shown in Figure 1. Grass playing surfaces (>99% probability), early season games (>91% probability), and higher temperature (>99% probability) games were all independently associated with a decreased risk of diagnosed concussion.

The point estimate for grass surface IRR was 0.78, meaning that for mid-season games (weeks 8 and 9) played at the average temperature for the dataset (~17 °C), the risk of concussion was ~22% lower on grass than on artificial turf. The 95% credible intervals (0.68, 0.89) indicated the true effect is likely between 11% and 32% reduction on grass compared with artificial turf.

Main effects for the week (IRR = 1.02 [1.00,1.04]) and temperature (IRR = 0.85 [0.76,0.95]), should be interpreted within the context of interaction effects. There was >75% probability that two-way interactions of surface × week influenced concussion risk (Figure 1B), with an estimated 1-week IRR of 1.02 (1.00, 1.04) on artificial turf and 1.04 (1.01, 1.06) on grass. In other words, for each 1-week change, there was a ~2% increased risk of concussions on artificial turf and ~4% increased risk on grass when the temperature was held constant at the average for the dataset. There was >84% probability that the two-way interaction of surface × temperature influenced risk. The point estimates for temperature IRR were 0.84 (0.76, 0.95) on artificial turf and 0.94 (0.86, 1.03) on grass. This translates to a ~16% reduced/increased risk of concussion on artificial turf and a ~6% reduced or increased risk of concussion on natural grass, for each standard deviation

TABLE 2. Summary of parameters from the primary negative binomial regression model, including relevant two-way interaction terms (n = 1830 games)

	β		Incidence Rate Ratio (IRR)	95% Credible Interval for IRR		Markov Chain Monte Carlo Diagnostics		
Parameter	Median	SD	Median	Lower (2.5%)	Upper (97.5%)	R-hat	Effective Sample Size (ESS)	Monte Carlo Standard Error (MCSE)
Grass Surface	−0.25	0.07	0.78	0.68	0.89	1.00	148,645	<0.001
Week	0.02	0.01	1.02	1.00	1.04	1.00	95,606	<0.001
Temperature Z-score	−0.17	0.06	0.85	0.76	0.95	1.00	88,739	<0.001
Grass × week	0.02	0.02	1.02	0.99	1.05	1.00	95,851	<0.001
Grass × temperature Z-score	0.11	0.07	1.11	0.96	1.28	1.00	86,924	<0.001

Grass surface was associated with a reduced risk of concussion compared with artificial surface (>99% probability). Concussion risk was also reduced with greater temperatures (>99% probability) and increased with each week of the season (>91% probability). There is reasonable probability (>75%) that each of the two-way interaction terms affects concussion risk. If the two-way interaction terms are interpreted as "no effect" and removed from the model, all three main effects have a >99% probability of influencing concussion risk (eAppendix: eTable 3, eFigures 6A,B and 7A,B; http://links.lww.com/EDE/C63). (Note, the lower bound credible interval for Week is reported as 1.00, but is 0.997, so the 95% credible interval does cross 1.0.)

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(corresponding to 7.9 °C) increase or decrease in temperature during the middle of the season.

The results indicate that the protective association of grass surface compared with artificial turf was somewhat attenuated in warmer conditions and late-season games (Figures 2 and 3). While the risk of concussion increases over the course of the season regardless of surface, this weekly increase in risk seems to be greater on natural grass than it is on artificial turf when the temperature is held constant (Figure 2). The risk of concussion seems to increase substantially on artificial turf in colder temperatures, whereas cold temperatures do not seem to exacerbate the risk of concussion on natural grass as much as holding the week constant (Figure 3).

Though the two-way interactions were likely associated with concussion risk (>75% probability), we also ran the models without these interactions to see if the results for the main effects remained stable. When this main effects model is used, all three independent predictors have a >99% probability (eAppendix: eTable 3A and eFigure 6A–C; http://links.lww.com/EDE/C63) and produced nearly identical point estimates to the primary model (eAppendix: eTable 3B; http://links.lww.com/EDE/C63).

Diagnosed Concussion Rates per Game

For an average temperature (~17 °C), mid-season game (week 8–9), the number of concussions predicted per game was 0.51 on natural grass and 0.66 on artificial turf. Early-season (week 1) warm-weather (25 °C) games have a concussion rate of 0.37 per game on natural grass and 0.48 per game on artificial turf. Late-season (week 16) cold-weather (5 °C) games have concussion rates of 0.72 and 0.96 per game, on the respective surfaces.

Sensitivity Analyses

Models investigating only indoor games or indoor and outdoor games in a moderate temperature range found similar IRRs for the week and their interaction with the surface as the main model (Table 2 and eAppendix: eTable 4B; http://links.lww.com/EDE/C63). A model investigating only games in weeks 13–16 found similar IRRs for temperature and its interaction with surface as the main model (Table 2 and eAppendix: eTable 5B; http://links.lww.com/EDE/C63). This confirmed both as independent predictors.

DISCUSSION

To our knowledge, this is the first study to comprehensively examine how the interaction of surface, temperature, and time in season factors interact to influence concussion risk, and the first to use a Bayesian approach to evaluating concussion risk. We found that the risk of diagnosed concussion

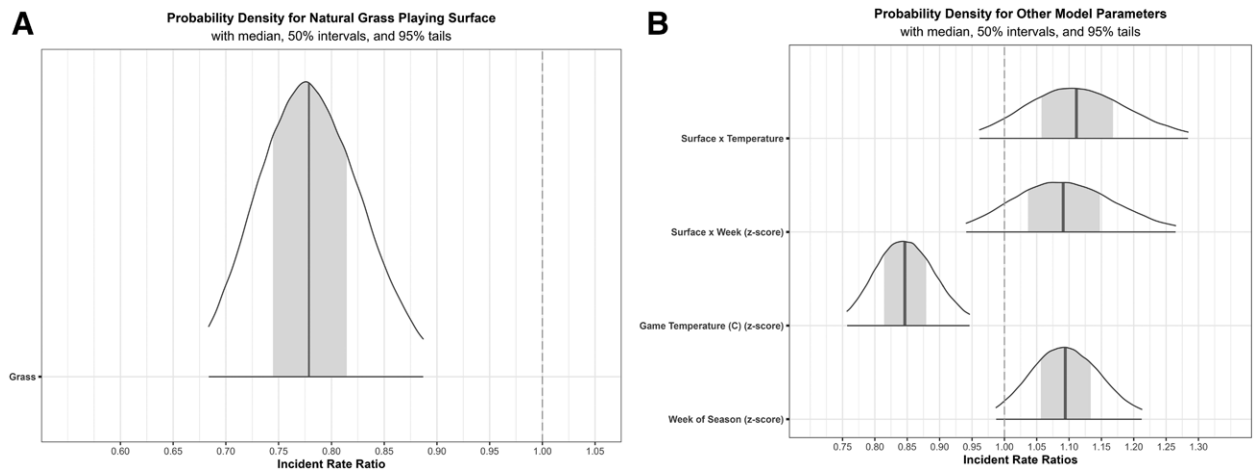

FIGURE 1. Probability distributions for all parameters primary model. All coefficients were exponentiated to compute the incidence rate ratio (IRR). For all figures, the vertical bar within each curve represents the median value for prior distribution, gray shading represents the 89% credible interval and the tails represent the 97% credible interval for each parameter coefficient. The intercept represents the concussion rate (number of concussions per game). The posterior distribution for the IRR of the natural grass surface is below 1.0 (vertical dashed line, for frame of reference), representing a decreased risk for concussion. A, Probability distribution for natural grass playing surface. The overall probability that concussion risk was reduced on natural grass compared to artificial turf was >98%, with a median incident rate ratio of 0.53 (~47% reduced risk). However, positive values for two-way interactions (B) suggest that the protective effect of natural grass may be diminished with greater temperatures and later in the season. B, Probability distribution for game temperature, week of season, and the two-way interactions included in the model. There was a high probability (>99%) that concussion risk was reduced at greater temperatures and increased later in the season (>91%). Much of the density of the posterior distributions for the week and the two-way interactions were >1.0, suggesting that these values may influence concussion risk, however cautious interpretation of these parameters may be warranted as their probabilities are a bit lower (grass × week: >75%; grass × temperature: >84%). If the two-way interactions are interpreted as "non-significant" and removed from the model, all main effect parameters increase to >99% probability of association with concussion risk (eAppendix: eTable 3; http://links.lww.com/EDE/C63).

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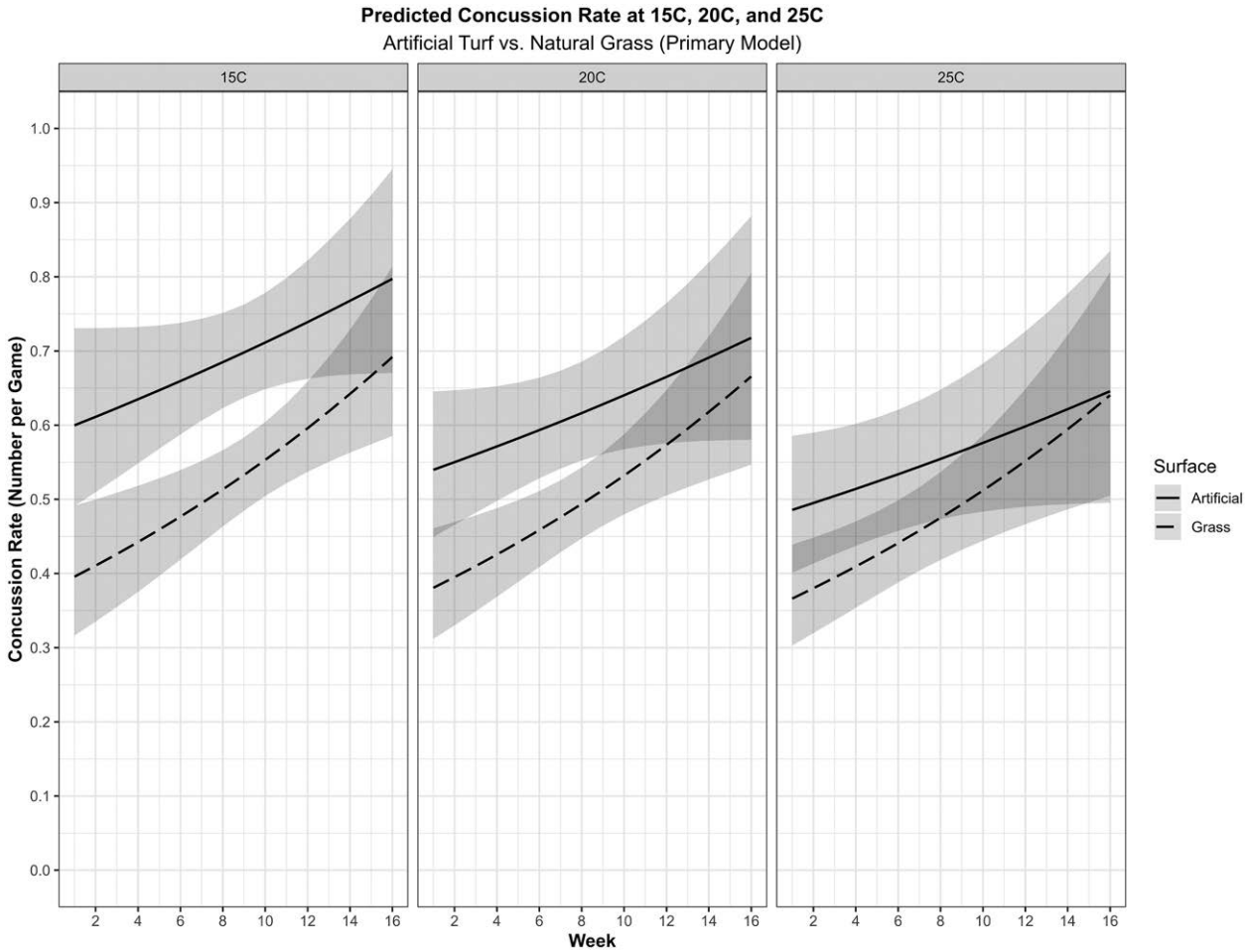

FIGURE 2. Marginal effects of concussion rate by the week of season, at a fixed temperature of 20 °C. The black spline (solid = artificial turf, dashed = grass) represents predicted concussion rate and gray shading represents the 95% credible intervals. These represent reasonably probable temperatures for outdoor games in temperate climates across the entire football season, and 20 °C represents the approximate temperature for indoor games (see eAppendix: eFigure 3A–C; http://links.lww.com/EDE/C63). At any given week of the season, the risk of concussion on artificial turf is magnified during colder temperatures.

is reduced on natural grass playing surfaces, increased later in the season, and increased in colder temperatures. While all of these factors independently modify diagnosed concussion risk, week and temperature both interact with the effect of playing surface. We posit that these factors modify the effect of playing surface (due to changes in surface material properties, described in *Explanatory Mechanisms* section), but it is also possible that playing surface modifies the effects of week and temperature. These findings provide compelling evidence that artificial turf is associated with greater concussion risk in the NFL and support the NFLPA's advocacy for transitioning to natural grass fields to improve player safety.[9] Additionally, these results are consistent with the hypothesis that extending the NFL season will likely increase concussion risk.

The large sample size combined with the Bayesian negative binomial regression model provides us with a high level of confidence in our results, which conflict with some existing literature with smaller sample sizes.[2] In an industry-funded study, Meyers found a slightly greater risk of concussion on natural grass compared with one specific manufacturer's (FieldTurf) artificial turf over three seasons of college football.[11] We ran a separate analysis comparing natural grass surfaces to FieldTurf surfaces and found natural grass was still associated with lower concussion risk (eAppendix: eTable 6A,B and Figure 8A,B; http://links.lww.com/EDE/C63). Discrepancies may be attributable to greater sample size in our study, differences between college versus professional football, or differences in the time period studied. The evidence that week and temperature interact with surface effects may also explain conflicting findings within the literature regarding the role of playing surface in athletic injuries.

Explanatory Mechanisms

The most likely explanation for lower concussion risk on natural grass compared with artificial turf is rooted in the biomechanics of helmet-to-ground impacts. A video review of

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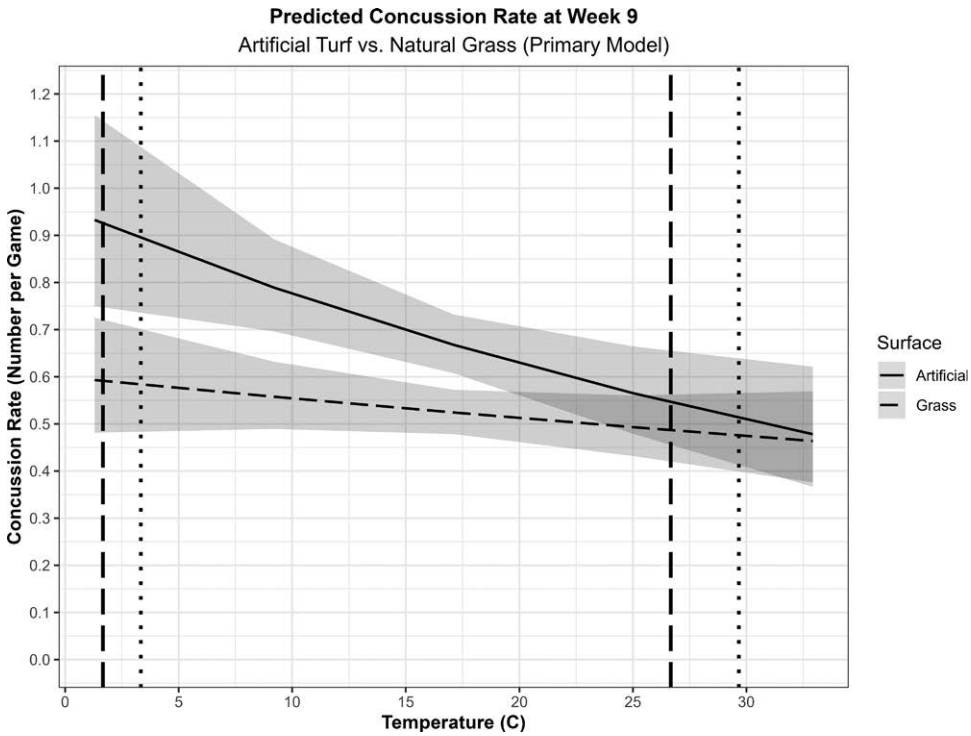

FIGURE 3. Marginal effects of concussion rate by temperature, fixed to week 9 of the NFL season. The black spline (solid = artificial turf, dashed = grass) represents predicted concussion rate and gray shading represents the 95% credible intervals. Week 9 is utilized for visualization because it represents a time of the season with a wide range of game temperatures. The vertical long-dashed and dotted lines represent the temperature extremes observed at week 9 within our dataset for artificial turf (n = 44 games, mean temperature = 13.4 °C) and natural grass (n = 57 games, mean temperature = 17.4°C), respectively. For games as cold as 1.5 °C, our model is quite certain that there will be more concussions on artificial turf than grass. At the other end of the temperature spectrum, however, our model is much more uncertain, as indicated by the substantial overlap in the credible intervals at temperatures above 20 °C. This overlap is to be expected due to the low sample size at higher temperatures at week 9, typically falls in the early part of November. For week 9 games, the average temperature in outdoor stadiums (n = 88 games, artificial turf and natural grass combined) is 15.0 °C (standard deviation = 6.8 °C).

injuries across five NFL seasons (1996–2001), suggests helmet-to-ground impacts account for 16%[39] to 22%[40] of concussions. A more recent study across two seasons (2015–2016) found that 18% of concussions with a clear source of impact were due to this mechanism.[41] The biomechanics of helmet-to-ground impacts are substantially different than player-to-player impacts, and football helmets are not tested (or designed) for these types of impacts.[42–44] These impacts are potentially more severe, due to unusually high angular velocities and accelerations, and the potential for rebounds (both of the head within the helmet and with the helmet and head combined).[43]

Grass tears in relation to translational force, whereas artificial turf does not, which allows the latter to have greater force transmission between the player and the surface.[45,46] Increased friction between the helmet and surface could lead to greater torque placed about the head upon a helmet–ground interaction.[43] Indeed, laboratory reconstruction of helmet-to-ground impacts has revealed that the severity of impact is highly sensitive to the compliance and frictional properties of the surface.[42] Reconstruction of 10 representative impacts causing concussions from NFL games demonstrated a greater severity index and head injury criterion for all helmet-to-ground impacts on artificial turf (n = 2) compared with helmet-to-helmet impacts (n = 8).[47] If our results are indeed driven by surface materials, this has implications for other athletes because head-to-ground impacts account for 5%–20% of concussions across various sports.[48–53]

Week of season and game temperature were independently found to influence concussion risk, which is also likely rooted in helmet impact biomechanics. Both surfaces degrade over the course of the season,[20,21] which can influence friction and energy absorption.[22,54] Likewise, surface temperature influences shock absorption properties of artificial turf,[23,24] and soil hardness and grass quality of soccer fields (moisture dependent, with the greatest effect in winter months).[54] Artificial surface temperature is also highly influenced by solar radiation,[55] which varies with the season. As demonstrated by the interaction effects (Figures 1C and 2A,B), there is a reasonably high probability that natural grass fields may also be prone to greater degradation within

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a season compared with artificial turf (e.g., soil compaction, substantial decrease in grass density, etc.),[54,56] and lose some of its apparent protective effects later in the season.

Various biologic factors could independently account for increased concussion risk later in the season. It has been theorized that a prior history of repeated sub-concussive head impacts could change one's biologic threshold for a concussion,[57] and there is mixed evidence to support this.[15–18,58–61] Additionally, cross-sectional data suggests that brain volume changes seasonally in the general population,[62] and this could theoretically influence concussion risk over the course of a season, regardless of game conditions or cumulative impact exposure.

It is also possible that these risk factors also influence an athlete's perception of concussion risk, concussion signs or symptoms, and/or voluntary reporting of concussion-related concerns. These factors may also influence medical personnel's vigilance in suspecting a concussion and/or accurately diagnosing it. For instance, if players or medical staff believe that artificial turf increases concussion risk, it is possible that they are more likely to suspect, report, or diagnose a concussion after a helmet-to-ground impact on artificial turf than a similar impact on natural grass. NFL players have expressed concerns about playing surface,[9] and elite soccer players also perceive tsurface influences injury risk,[63] and some even believe this is temperature-dependent.[64] However, it is unknown how these perceptions influence self-recognition of injury symptoms and reporting to medical staff. It is also possible that athletes may be more or less likely to report symptoms at different points of the season and may also be influenced by high-stakes competitive outcomes.[65] Indeed, NFL players diagnosed with a concussion are more likely to face salary reductions and be released from their team.[66]

Limitations

Concussions are underdiagnosed across various sports and proficiency levels, so it is possible that some of these injuries were missed and our analysis cannot provide any insight on those. Players are flagged by medical staff to be evaluated for concussion based on high-risk situations and visible clinical signs (e.g., ataxia, loss of consciousness, and slow to get up),[67] but these are not as common as headache or dizziness (reported by ~55% and ~41% of documented concussions in NFL players, respectively).[35] Analysis of the 2017 NFL season video footage revealed ~26% of players diagnosed with concussions did not exhibit visible signs, and thus required more thorough medical evaluation.[68] This means that concussion recognition or diagnosis is highly (but not entirely) dependent on honest reporting from players. Survey research has revealed that ~23% of collegiate football players did not intend to report concussion symptoms (with another 9.6% uncertain if they would),[69] but concussion nondisclosure data from modern-day NFL players are not available. Thus, the data used in this study (and much of the sports-related concussion epidemiology literature) only represent documented or diagnosed concussions. Therefore, we are confident that our model identifies risk factors for diagnosed concussions, but it is unknown if it is representative of all concussions (i.e., documented and undocumented). This would likely be the case if these risk factors were purely physical or biological, but it is possible that concussion reporting and diagnosis are biased in either direction by some of these risk factors (as described in the previous section).

There is also considerable heterogeneity within the broad categories of artificial turf and natural grass,[46] and we did not attempt to differentiate between them (with the exception of Field Turf, eAppendix: eTable 6A,B; http://links.lww.com/EDE/C63). In other sports, certain species/cultivars may be associated with differing risks of musculoskeletal injuries independent of ground hardness (attributable to differences in thatch),[70,71] though not concussions.[71] Surface moisture can also influence surface material properties.[72] One could attempt to account for this by including weather in analysis; however, preexisting precipitations influence the surface even if game-time weather is dry, and weather can vary considerably within a game. We were also unable to quantify field "wear" given different environmental conditions (e.g., sunlight) and nongame activity (e.g., concerts, setup, and maintenance),[73] and these may vary between stadia.

Given the increased risk of concussion later in the season and at colder temperatures, we suspect that the concussion risk will be greatest during the playoff season. However, accurate injury reports are not available for teams that fail to advance to the next round of playoffs (or after the championship game), which precludes the possibility of a valid analysis of concussion risk during this time.

CONCLUSIONS

Eight years of data demonstrated that diagnosed concussion risk is substantially increased in NFL games played on artificial turf. While concussion risk also increases with cold weather, regardless of the playing surface, the risk is particularly amplified on artificial turf. There is also an increased risk of concussion later in the season, even when temperature is controlled for, but it remains uncertain if this is due to biologic factors, field degradation, or a combination. These data suggest that extending the NFL season may provide greater health risk to players and that player concerns over the use of artificial turf are warranted. Further research is necessary to determine if these findings extend to nonprofessional levels of American football and other team sports.

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